

Research article

Modeling and prediction of climate change impacts on water resources vulnerability: A multi-model approach

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ABSTRACT

In a rapidly changing world, uncontrolled climate change worsens water scarcity disrupting hydrological cycles and hindering sustainable development. Addressing water resources vulnerability requires holistic approaches to better understand complex systems, mitigate risks from changing weather patterns, and develop adaptive water management strategies. In this study, we modeled climate change impacts on water resource vulnerability using machine learning (ML) and SWAT model based on CMIP6 Global Climate Model (GCMs) under Shared Socio-economic Pathway (SSP). Six ML models were evaluated to reliably predict hydroclimatic events; Extremely Randomised Trees (ERT) and Categorical Boosting (CatBoost) performed best for simulating ensemble climate interactions. The statistical indicators confirmed model reliability reducing input uncertainties with bias-corrected datasets. The ensemble SWAT model simulation showed a good agreement between simulated and observed values ($R^2 = 93\%$, $NSE = 91\%$, and $PBIAS = -1.08\%$) for calibration and ($R^2 = 94\%$, $NSE = 93\%$, and $PBIAS = -2.32\%$) for validation periods. Furthermore, we developed a novel Hydrologic Vulnerability Index (HVI) framework based on water balance components to quantify watershed vulnerability dynamics across baseline and future scenarios. The HVI ranged from low to extreme, with maximum lower values (54.03 %) observed at baseline, indicating resilience to hydrological stress, and higher values indicating severe vulnerability (43.45 %) at SSP245, indicating extreme drought conditions. The HVI framework integrates climate projections with actionable insights, offering a comprehensive approach to sustainable water management, adaptive infrastructure, and targeted interventions. Hence, innovative policies are critical to address extreme HVIs ensuring resilience against water scarcity and ecosystem degradation. This study underscores the importance of coupling data-driven hydrological analysis with climate responsiveness for effective watershed and environmental sustainability. These results demonstrate the importance of integrating various perspectives and strategies to address both short- and long-term climatic problems, by employing adaptive management practices to ensure sufficient water and ecosystem resilience.

1. Introduction

Climate change threatens food security, drinking water, and energy use, highlighting the need for adaptation and building resilience (IPCC, 2014). Climate change affects the vulnerability of watersheds to hydrologic imbalance, while anthropogenic activities aggravate water stress and hinder sustainable development (Conway and Schipper, 2011; IPCC, 2007; Mahdian et al., 2023; Sillmann et al., 2017). It has unpredictable implications for the availability and quality of water resources, complicating water management due to uncertainty (Lespinas et al., 2014; Wilby and Harris, 2006). As temperatures rise and precipitation

patterns evolve, both human populations and ecological systems in water-scarce regions are highly affected (Zscheischler et al., 2018). Fluctuations in anthropogenic forcing aggravate droughts and floods (Liu et al., 2017; Rodell et al., 2018), and water quality problems (Meresa et al., 2022; Milly et al., 2005), and affect agricultural water supply (Moon et al., 2024; Zhao et al., 2023). Streamflow responses to climate change at the watershed level are crucial for water resource sustainability and ecosystem functionality (IPCC, 2021; Ma et al., 2022; Raymond et al., 2020). Hence, evaluating regional climate change impacts on water resource is vital for developing effective management strategies (Cuthbert et al., 2019; Marin et al., 2020).

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Global Climate Models (GCMs) are essential for understanding climate change; however, their coarse resolution makes them unsuitable for regional adaptation plans (Zelinka et al., 2020). Though GCM with high resolution require downscaling for watershed studies, they are widely used to estimate future climatic scenarios (Gebrechorkos et al., 2023). Downscaling and bias correction methods used to refine GCM outputs to address local and regional needs (Rampal et al., 2024; Wang et al., 2004). Regional Climate Models (RCMs) incorporates local factors, though it is computationally intensive and requires a large datasets (Ahmed et al., 2013; Gudmundsson et al., 2012; Teutschbein and Seibert, 2012). Bias correction addresses systematic errors in the model outputs (Ehret et al., 2012). Model simulations can be adjusted to account for significant biases enhancing climate change effect estimates (Babaousmail et al., 2022; Kesavavarthini et al., 2023). Climate policy and decision makers face challenges owing to biased data uncertainties and disregarding these issues can lead to inadequate climate prediction and adaptation (Lafferty and Srivier, 2023; Mahony et al., 2022). Selecting correct GCM and bias correct based on how well it reproduces is important (Gao et al., 2023; Hakala et al., 2018).

The Sixth Phase Scenario-Based Coupled Model Intercomparison Project (CMIP6) (Eyring et al., 2016b), stressed GCMs are indispensable, providing better physical representations and future assessments incorporating various Shared Socioeconomic Paths (SSPs) (Gidden et al., 2019; Riahi et al., 2017; van Vuuren et al., 2017). Although CMIP6 has uncertainties, combining physical models with GCM improves output reliability to address sustainability and management issues (Yu et al., 2023). While CMIP6 has improved the atmospheric process representation, it still has flaws, making it challenging to use easily (Londhe et al., 2023; Taye et al., 2011). Though computationally expensive, ensemble modeling is usually more efficient and reliable than single-model simulations for reducing model uncertainty (Kysely et al., 2011; Milinski et al., 2020; Mondal et al., 2021).

Watershed vulnerability to erratic climate change requires thorough hydrological assessments (Teutschbein and Seibert, 2012; Yang et al., 2021). The Soil and Water Assessment Tool (SWAT) is a commonly used watershed model (Arnold et al., 1998). However, the quality of input uncertainty often complicates risk management (Joseph et al., 2018). Uncertainties in hydrological modeling complicate sustainable water resource planning and climate simulation (Mengistu et al., 2021). Hence, further studies are required to understand the impact of climate change on watershed vulnerability for effective climate adaptation. To manage water resources and respond to climate change, a supplementary modelling strategy is needed. Combining machine learning model (ML) and SWAT models improves climate change projections on basin hydrology. ML captures nonlinear correlations and complex climatic

dynamics, identifying hidden patterns to predict watershed responses to climate change (Ahmed et al., 2024; Miao et al., 2024; Ullah et al., 2023; Wu et al., 2023). Hence, the main objective of this study was to assess the impacts of climate change on water resource vulnerability, coupling ML and SWAT model. Furthermore, we developed a novel spatiotemporal hydrologic vulnerability index (HVI) for the first time based on enhanced coupled simulations. HVI evaluates a watershed's risk to hydrological stress and water resource challenges.

2. Materials and methods

2.1. Datasets

The vulnerability assessment relied on various data sources to estimate driving hydrological processes. The daily time-resolution observed climate data of Upper Gilgel Gibe watershed in Ethiopia (Fig. 1) from Jimma, Shebe, Assendabo, Busa, Dedo, and OmoNada stations were used for hydrologic ensemble simulation. Precipitation (mm) and minimum and maximum temperatures ($^{\circ}\text{C}$) were extracted from grid-based General Circulation Models (GCMs) of official Earth System Grid Federation (ESGF). The anthropogenic forcing for simulation of climate-induced hydrological changes was derived from Coupled Model Intercomparison Project Sixth Phase (CMIP6) GCMs project under Shared Socioeconomic Paths (SSPs) (Riahi et al., 2017). The historical GCMs simulation outputs driven by six CMIP6 GCMs, CanESM5 (Canadian), MIROC6 (Japan), MRI-ESM2-0 (Japan) (Oshima et al., 2020), IPSL-CM6A-LR (France), EC-EARTH3 (Japan) (Döscher, 2021), and GFDL-ESM4 (USA), were used as the initial boundary conditions (r1i1p1f1) (Eyring et al., 2016a). To explore potential changes in hydrology, we used GCMs representing CMIP6 under SSP245 (moderate scenario), SSP370 (scenario with higher challenges to mitigation and adaptation), and SSP585 (high-emission, fossil-fuel-intensive scenario) (Gidden et al., 2019). Historical climate data are crucial for influencing hydrological processes; however, climate model data may diverge from observed data, and these biases must be corrected prior to impact assessments of climate scenarios.

2.2. Bias adjustment

Transforming coarse-resolution GCM outputs into data that are applicable using the GCM as boundary conditions for model simulations. Bias corrections augment hydrological modeling prediction reliability to address inherent biases in GCM outputs by reducing systematic errors (Lafferty and Srivier, 2023; Supharatid et al., 2022). Quantile Mapping (QM) aligns the cumulative distribution functions of precipitation data

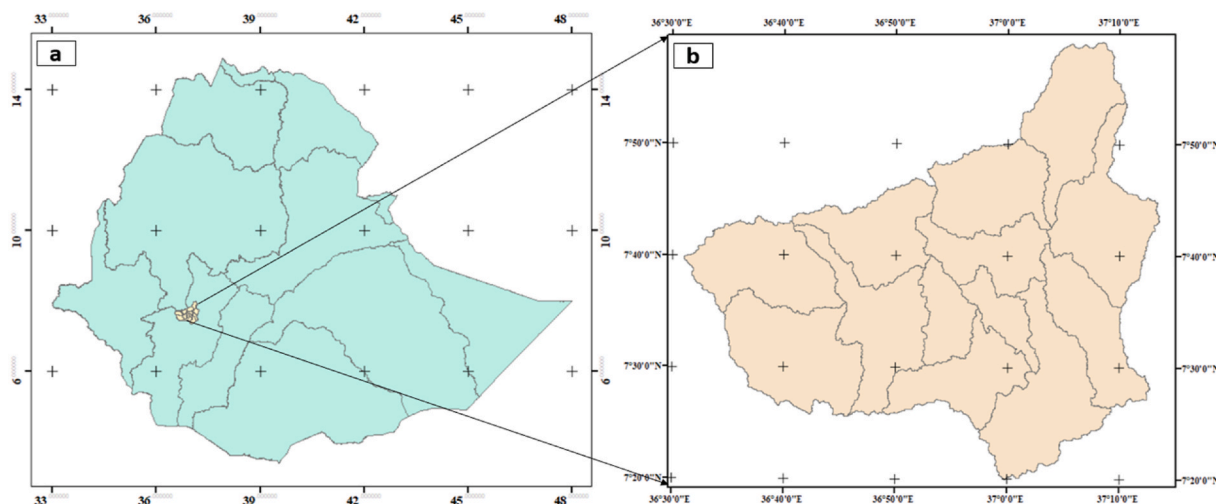


Fig. 1. Location of the study area: a). Boundary of Ethiopia, b). Upper Gilgel Gibe watershed.

to improve the representation of means and extremes (Rajczak et al., 2016), whereas Variance Scaling (VS) adjusts modeled temperature variances to reflect observed variability to predict extreme temperatures accurately (IPCC, 2007; Maraun and Widmann, 2018), ensuring refined outputs align with observed patterns (Ehret et al., 2012). QM corrects precipitation (mm) discrepancies, whereas VS capture variability in maximum (Tmax) and minimum (Tmin) temperature (°C) before data assimilation to reduce uncertainties (Kim et al., 2022; Piani et al., 2010). Hence, bias correction of CMIP6 GCMs was extracted from historical data (1988–2014) to enhance future projections (2015–2100) reducing systematic errors to ensure accurate inputs of SWAT model simulation. The overall study design flowchart is as shown (Fig. 2). The subsequent evaluation used statistical metrics such as Percent Mean Bias (PBIAS), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Pearson Correlation Coefficient (r) between the CMIP6 GCMs of precipitation, Tmax, and Tmin, and observation data. Mathematically, these metrics can be calculated (Eqs. (1)–(4)).

$$PBIAS = \frac{\sum_{i=1}^n (Si - Mi)}{\sum_{i=1}^n (Mi)} \quad (1)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Si - Mi)^2} \quad (2)$$

$$r = \frac{\sum_{i=1}^n (Si - Sm)(Mi - Mm)}{\sqrt{\sum_{i=1}^n (Si - Sm)^2} \sqrt{\sum_{i=1}^n (Mi - Mm)^2}} \quad (3)$$

$$MAE = \frac{\sum_{i=1}^n |Si - Mi|}{n} \quad (4)$$

Where Si and Mi are the simulated and observed values of the climate variables, respectively, i refers to the simulated and observed pairs, and n is the total number of pairs.

2.3. SWAT model

The Soil and Water Assessment Tool (SWAT) physically based, semi-distributed hydrological model simulates water balance (Arnold et al., 1998). The historical SWAT model simulation was based on ensemble GCM for reliable hydrological balances. The SWAT calibration and uncertainty procedure (SWAT-CUP), using the Sequential Uncertainty Fitting 2 (SUFI-2) method used to achieve successful model calibration (Abbaspour et al., 2004). The automatic calibration technique operates within predefined time constraints, using realistic parameters based on (Mengistu et al., 2019). Then, assessed using metrics such as the

Nash-Sutcliffe efficiency (NSE), coefficient of determination (R^2), and Percent Bias (PBIAS), measuring the tendency of simulated and observed (Gupta et al., 1999).

2.4. Machine learning

Accurate weather prediction is crucial for understanding water resource availability, flood risks, droughts, and water infrastructure management (Ahmed et al., 2024). Artificial Intelligence (AI) models have been increasingly applied to water resources research to capture complex nonlinear relationships between variables of large datasets (de Burgh-Day and Leeuwenburg, 2023). Machine learning (ML), a subset of AI, transforms vast climate datasets into actionable prediction patterns (Asadollah et al., 2022). In this study, the most widely adopted ML has been implemented for training and prediction purposes (Miao et al., 2024; Zounemat-Kermani et al., 2021), to address hydrological challenges.

Random Forest (RF) is collects numerous decision trees to boost prognostic accuracy and reduce overfitting (Breiman, 2001). To enhance robustness, each decision tree is trained on a random subset of data, minimizing variance and reducing errors (Fernández-Delgado et al., 2014). RF has widely been implemented in hydrological applications offering insight into climate data features interpretability (Biau and Scornet, 2016; Wu et al., 2023). Decision Trees (DT) are flexible ML used for classification and regression tasks, characterised as a flowchart, where internal nodes resemble feature splits, divisions indicate decision rules, and leaf nodes indicate results (Quinlan, 1986). DT is known for its interpretability, model complex nonlinear interactions handling numerical data (Hastie et al., 2009; Loh, 2011). Extremely Randomised Trees (ERT) are an ensemble ML that introduces extra randomness by selecting feature splits and defining split points randomly, thereby enhancing generalisation and reducing overfitting (Geurts et al., 2006). ERT builds multiple DT independently and then averages them to reduce variance and bias.

Extreme Gradient Boosting (XGBoost) is a powerful ML highly efficient in handling structured data, and assimilate regularization to reduce overfitting (Chen and Guestrin, 2016). XGBoost uses Shapley Additive Explanations (SHAP) to identify factors influencing hydrological simulations, optimize accuracy and adaptable to various applications (Wang et al., 2022). A Light-Gradient Boosting Machine (LightGBM) is a gradient-boosting algorithm developed to improve performance. LightGBM reduces computational time by refining the extrapolative power while alleviating dimensionality issues and supporting categorical features (Bentéjac et al., 2021; Ke et al., 2017). Categorical Boosting (CatBoost) automatically processes data using effective gradient boosting to reduce leakage during training (Dorogush et al., 2018; Hancock and Khoshgoftaar, 2020; Prokhorenkova et al., 2018), and has been effectively employed in hydrological and climate applications (Gibson et al., 2021).

RF combines decision trees to predict climate variables based on temperature and precipitation under various emission scenarios (SSP245, SSP370, SSP585). It reduces overfitting through key parameters like the number of trees, tree depth, and bootstrap sampling. DT are simpler models that split data based on feature values, influencing performance through parameters like tree depth and sample requirements. ERT, a variation of RF, introduces additional randomness in tree construction to improve robustness and reduce bias in predictions. XGBoost and LightGBM are efficient gradient boosting methods for climate predictions, focusing on parameters like learning rate, tree count, and maximum depth, with LightGBM particularly suited for large datasets. CatBoost, designed for handling categorical data, also uses climate inputs to predict future variables, excelling with complex datasets.

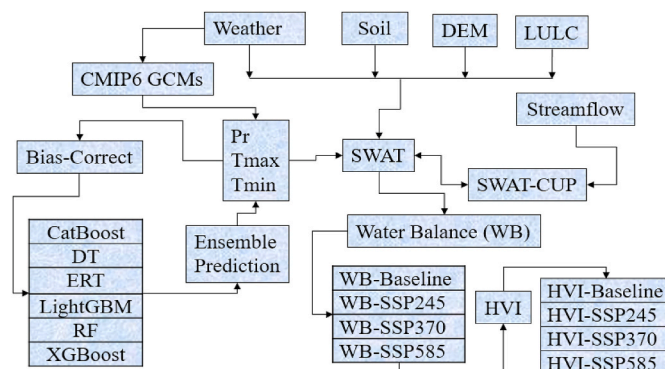


Fig. 2. Schematic overview of the methodology.

2.5. Ensemble hydrologic modeling

The effects of temperature and precipitation patterns on hydrology were assessed using GCMs based ML models. Bias-corrected GCMs and their ensembles were used for evaluating model in reproducing the watershed hydrology with observed baseline climate (1988–2014) and project future climate change (2015–2100). To address associated uncertainties in the projected climate, a multi-model ensemble bias-corrected GCMs were used for projecting climate variables (Milinski et al., 2020; Zounemat-Kermani et al., 2021). An ensemble of GCMs allows ML approach to achieve better results than individual models for assessing future watershed hydrology and ecosystem health. We simulated hydrological mechanisms of future periods (2015–2044, 2045–2074, and 2075–2100) using ML models to force hydrological models.

2.6. Hydrologic vulnerability assessment framework

We propose a novel framework for developing Hydrologic Vulnerability Index (HVI) to assess water resource resilience by integrating hydrologic variables (water yield (WYLD), sediment yield (SYLD), surface runoff (SURFQ), groundwater flow (GWQ), precipitation (Pr), evapotranspiration (ET), potential evapotranspiration (PET), percolation (PERCO), and lateral flow (LATQ)). HVI helps to identify ecologically sensitive areas to sustain biodiversity and ecosystem health by sustaining environmental flow requirements. To standardise and quantify the relative importance of each variable, we categorised and weighted them based on the thresholds of water resource stress. A vulnerability score was computed for each grid cell and normalized using linear transformation to produce HVI for baseline and future projections (SSP245, SSP370, and SSP585) across the region. The proposed equation for calculating the HVI of each grid cell is given in Eq. (5).

$$HVI = \sum_{i=0}^n (R_i) \times (W_i) \quad (5)$$

Where W_i is the weight assigned to each hydrological variable, n is the maximum number of variables, and R is the reclassification score of each hydrological variable.

3. Results and discussion

3.1. Global climate models performance

This study presents detailed climate models reliability assessment with statistical performance metrics at diverse geographic stations. IPSL-CM6A-LR and CanESM5 had lower RMSE and MAE values, whereas GFDL-ESM4 and MIROC6 had higher errors (Fig. S1a). The IPSL-CM6A-LR had r values of 0.9850 and 0.9932 for the Assendabo and Shebe stations, respectively, indicating strong agreement with low RMSE and MAE and good accuracy and low prediction error (Fig. S1a). IPSL-CM6A-LR indicated moderate PBIAS. GFDL-ESM4 had a moderate RMSE and large r , indicating strong accuracy, whereas a higher PBIAS indicates projection bias. Despite the good RMSE and r values, MIROC6 had high PBIAS. The ensemble model, which combined all model outputs, indicated a significant reduction in prediction error, suggesting high agreement with observed precipitation. This highlights potential of ensemble model to reduce individual model biases improving precipitation prediction accuracy for robust climate simulations.

The IPSL-CM6A-LR demonstrated a strong Tmax simulation in terms of RMSE and MAE across all stations, reflecting its minimal error in predicting Tmax (Fig. S1b). A larger PBIAS, particularly in OmoNada, demonstrated a significant bias. The moderate PBIAS for IPSL-CM6A-LR suggests that it maintains a bias that does not significantly undermine overall performance. Despite being strong in terms of metrics, MIROC6

revealed a higher RMSE and MAE than IPSL-CM6A-LR and ensemble. The RMSE and r of MIROC6 for Busa indicated a lower Tmax prediction (Fig. S1b). The ensemble model, with a low RMSE and MAE, confirms its excellent performance.

In Tmin simulation, IPSL-CM6A-LR had feasible RMSE at Assendabo and high r ; but, higher PBIAS. The ensemble model incorporated minimum biases, attaining lowest MAE and RMSE for all stations, and largest at Busa and Jimma (Fig. S1c). MIROC6 predicted Tmin less accurately at OmoNada because of its lower r value (Fig. S1c). The ensemble Tmin, RMSE, and MAE were reduced, thereby improving the accuracy. The IPSL-CM6A-LR and CanESM5 managed better, whereas MRI-ESM2-0 had higher RMSE. In contrast, MRI-ESM2-0 and MIROC6 were inconsistent. At Assendabo, MRI-ESM2-0 had a significant RMSE, indicating a prediction error for Tmin (Fig. S1c). Although MIROC6 outperformed MRI-ESM2-0, it exhibited a higher RMSE at several stations than the ensemble and IPSL-CM6A-LR. Hence, the ensemble model, indicating better temperature prediction, is robust and suitable for climate simulations.

The differences in climate model performance can be attributed to factors such as model structure, parameterization, input data quality, and prediction consistency. Higher-performing models like IPSL-CM6A-LR and CanESM5 exhibit lower RMSE and MAE values, suggesting more refined parameterization and a better representation of climate variability. In contrast, lower-performing models like MIROC6 and MRI-ESM2-0 show higher errors and biases, indicating limitations in parameter tuning and input data reliability. IPSL-CM6A-LR demonstrates strong consistency across Tmax and Tmin simulations, whereas MIROC6 struggles, particularly with Tmin predictions. The presence of moderate to high PBIAS in some models highlights systematic biases that affect accuracy. Mainly, the ensemble model significantly reduces prediction errors by integrating multiple outputs, improving agreement with observed data. This suggests that while individual models have inherent biases, ensemble approaches enhance the robustness and reliability of climate simulations.

3.2. Machine learning model performance

Robust ML predicts climate patterns were evaluated using six ML for their efficacy (Fig. 3), characterizing potential impacts of emission scenario gradually increasing severity. For SSP245, RF was less accurate than ERT and CatBoost. For SSP245, DT had high error rates of MAE and RMSE (Fig. 3) which suggests that DT cannot accurately predict precipitation patterns in the region. RF, XGBoost, and LightGBM may be effective; however, ERT and CatBoost perform better (Fig. S2a; Fig. 3). SSP370 (Fig. S2b) and SSP585 (Fig. S2c), ERT demonstrated remarkable performance, with low MAE and RMSE and high R^2 (Fig. 3). For SSP245 (Fig. S2a), CatBoost provided accurate predictions, although it was slightly outperformed by ERT (Fig. 3). For SSP585 (Fig. S2c), CatBoost showed a slight increase in error; however, it remained a reliable model for precipitation prediction across all scenarios, making it a valuable alternative. In SSP370 (Fig. S2b), ERT performed better, with low MAE and RMSE values (Fig. 3). With a low MAE and high R^2 , ERT was able to sustain changes in precipitation even during the most extreme SSP585. Compared with ERT and CatBoost, which surpass RF in terms of precision, XGBoost and LightGBM fall apart. The XGBoost and LightGBM models performed poorly with more errors.

For SSP245, DT yielded high error rates (Fig. 3) and had largest errors, unable to capture the complex Tmax dynamics. RF demonstrated low errors and high R^2 in SSP245 (Fig. 3), declined slightly as severity increased. For SSP245, XGBoost performed better than RF; however, XGBoost performed worse than CatBoost and ERT (Fig. 3). In SSP245 (Fig. S3a), CatBoost demonstrated very low errors, indicating a high variance (Fig. 3). As severity worsened to SSP370 (Fig. S3b) and SSP585 (Fig. S3c), CatBoost resulted in low error and high R^2 . LightGBM shows strong performance, but slightly lags behind CatBoost and ERT in capturing significant Tmax variability, especially for highly sensitive

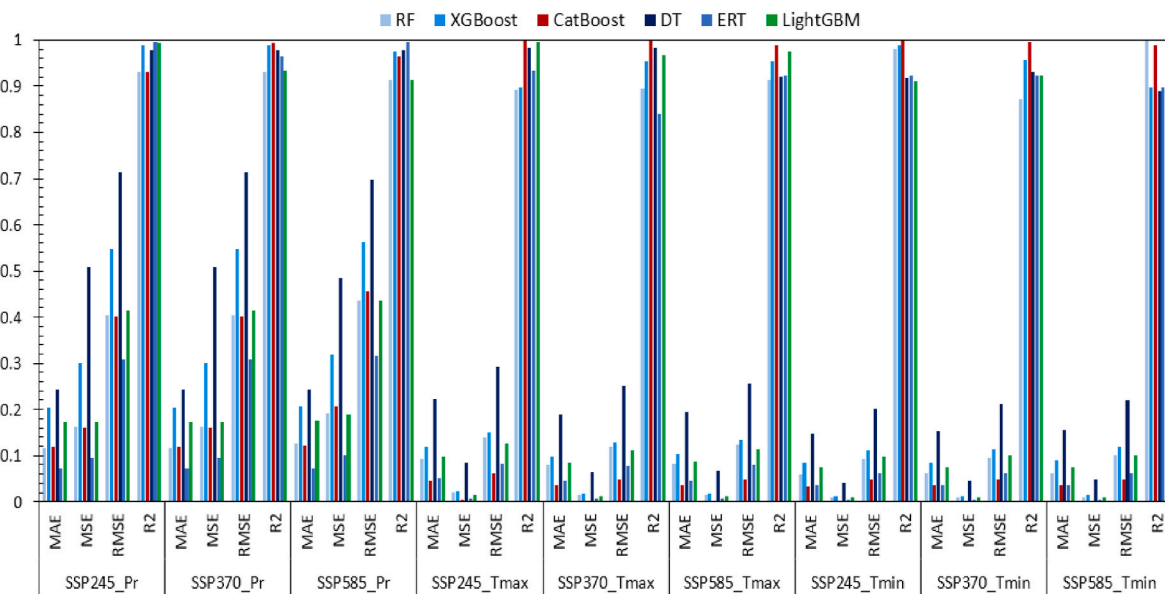


Fig. 3. Machine learning model performances indices under SSPs.

flood forecasting or drought predictions. Although ERT showed a strong performance, particularly with a high R^2 , CatBoost slightly outperformed it in error metrics, making it best choice for Tmax prediction. Accordingly, CatBoost (Fig. S3) was selected as a reliable model.

RF provides good Tmin performance for SSP245 (Fig. 3), but is not better than XGBoost, CatBoost, and ERT. XGBoost performed well but had higher errors than CatBoost and ERT. CatBoost consistently outperformed other ML while maintaining low errors (Fig. 3; Fig. S4a). Across SSP370 (Fig. S4b) and SSP585 (Fig. S4c), CatBoost and ERT had a low MAE and high R^2 (Fig. 3). LightGBM executes well but has a higher error than CatBoost or ERT, making it less suitable. ERT is also a strong candidate for Tmin predictions, with a slightly higher error than CatBoost (Fig. 3). DT performed poorly in SSP585, indicating weak generalisation, unreliable in the region. Hence, CatBoost (Fig. S4a–c), was the most reliable model for accurately predicting Tmin, even when climatic conditions became more extreme, providing robust prediction of temperature extremes.

Overall, compared to XGBoost and LightGBM, which rely on gradient-based optimization, ERT and CatBoost were identified as the top-performing models due to their higher accuracy, robustness, and ability to handle complex climate data. ERT demonstrated consistently low MAE and RMSE along with high R^2 , particularly under extreme climate conditions due to its extra-randomized splits, allowing to sustain accuracy as climate severity increases. Extreme randomization minimizes variance and enhances performance on highly variable climate data. CatBoost, consistently delivered low errors and high R^2 across all climate variables, making it particularly effective for predicting Tmax and Tmin. Hence, ERT and CatBoost outperform other models by effectively capturing complex climate patterns while maintaining high accuracy and generalization across varying emission scenarios.

3.3. Inferences of ensemble modelling

Climate projections offer insights into future climatic conditions and their implications for water resources. The CMIP6 GCMs multi-model ensemble provided enhanced average and extreme events, helping to better predict, mitigate, and make decisions. CatBoost is the most suitable ML for Tmin prediction under various scenarios accurately reflecting intricate interactions between climatic variables. ERT predicted SSP245 and SSP585 precipitation better than CatBoost (SSP370), offering considerable flexibility, while CatBoost dominated temperature

extremes suggesting that can be valuable for predicting heatwaves. The higher ERT performance to predict precipitation (SSP245 and SSP585) highlights its role in flood risk assessment. Although time-consuming and computationally expensive, we found that ensemble modeling is suitable for projecting climate variables, and is more reliable than single-model simulations, as supported by (Asadollah et al., 2022; Zounemat-Kermani et al., 2021). ML provides accurate predictions for mitigating the impacts of climate change, addressing potential flooding and drought, ensuring sustainable water availability, and preserving ecosystem health.

Overall, Ensemble modeling offers significant advantages over single-model simulations by reducing biases and uncertainties, improving the robustness of climate projections, and providing better regional climate predictions. Unlike individual models, which may have structural and parametric limitations, ensembles average across multiple models, minimizing errors and improving reliability. They also capture a wider range of internal climate variability, leading to more complete projections of future climate situations. Furthermore, it enhances confidence in future climate projections by accounting for different model structures and emission scenarios, making the results more resilient and robust.

3.4. SWAT model evaluation

The SWAT model was calibrated using the observed streamflow to show that the ensemble simulation of watershed hydrology is representative. The calibration parameters and optimised values used were GW_DELAY = 140.230 for groundwater delay, GWQMN = 0.342 for the shallow water aquifer threshold depth, and CN2 = 0.078 for the moisture condition II curve number. We measured soil depth (SOL_Z = 0.681), groundwater revap coefficient (GW_REVAP = 0.070), and surface runoff lag coefficient (SURLAG = 12.744). The soil evaporation compensation factor (ESCO = 0.052), saturated hydraulic conductivity (SOL_K = 0.253), and available water capacity (SOL_AWC = 0.223). Plant uptake compensation factor (EPCO = 0.705), coefficient of depletion of groundwater (ALPHA_BF), and maximum canopy storage (CANMX = 75.850). As shown (Fig. 4), hydrograph for both calibration and validation periods demonstrated that the observed and predicted values agreed. Furthermore, model performance was assessed with R^2 , NSE, and PBIAS values of 0.93, 0.91, and -1.08 for calibration and 0.94, 0.93, and -2.32 for validation periods, confirming predicted values

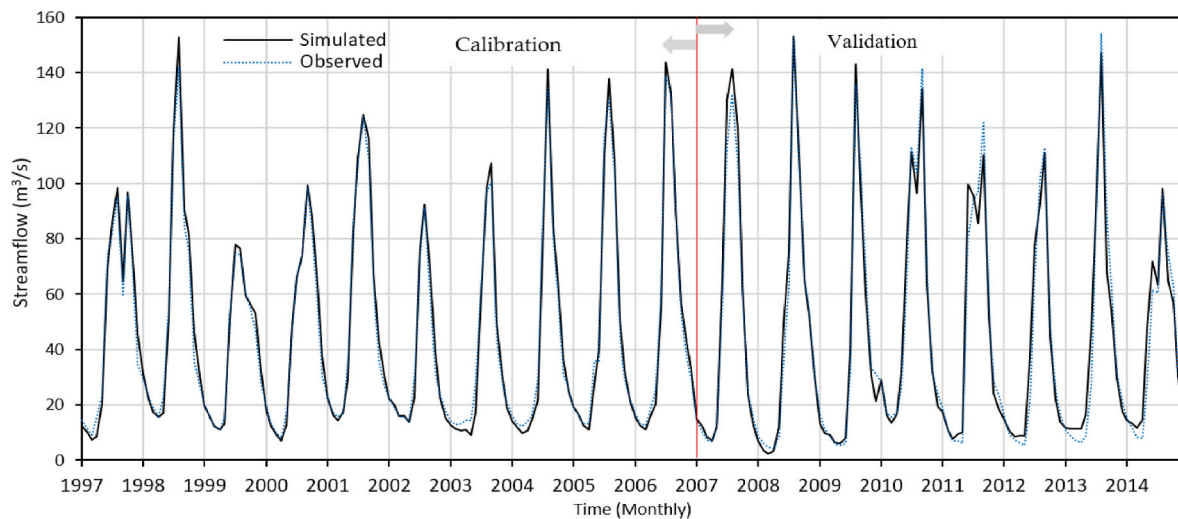


Fig. 4. Streamflow hydrograph for SWAT model ensemble simulation.

corresponded to observations as in (Moriassi et al., 2015).

The improvements in model performance under ensemble simulation are evident across all indices. The R^2 increased by 5 % in calibration and 9 % in validation, showing that the corrected model explains a greater proportion of the observed variance. The larger improvement in validation underscores the model's stronger predictive capability for unseen data, reinforcing its reliability for real-world applications. Similarly, the NSE improved by 4 % in calibration and 8 % in validation, reflecting an increase in accuracy relative to the mean of observations. This confirms that the ensemble simulation reduces errors more effectively in independent datasets, a critical factor for robust predictions. The most notable reduction occurred in PBIAS, which decreased by 6.82 % in calibration and 1.68 % in validation. The drop in calibration implies a significant reduction in systematic over- or under-prediction. The improvement in validation shows balanced simulations, as PBIAS directly influences decision-making process in water management enhancing the reliability of model outputs. Overall, the ensemble model demonstrates 5–9 % better explanatory power (R^2), 4–8 % higher predictive accuracy (NSE), and a 6.82 % reduction in calibration bias (PBIAS). The validation metrics confirm its generalizability, ensuring greater confidence in applications for climate, hydrological, and environmental management. Overall, we confirmed that ensemble GCMs simulated SWAT model better than based solely on observed variables (Mengistu et al., 2022). This suggests that ensemble GCMs can complement existing frameworks by enhancing model predictions, thereby reducing input uncertainties and providing more realistic estimates of hydrological responses to support water resource planning and management.

3.5. Watershed hydrological responses

The ensemble historical simulation demonstrated significant variability, peaking in July and August when precipitation surpassed 200 mm. The PET (mm) is the maximum possible loss of water to the atmosphere under optimal conditions, and demonstrated relatively consistent patterns, peaking in March (119.87 mm). Understanding PET is crucial for anticipating the water demand for crops, mainly during dry periods, when actual evapotranspiration may exceed available water. Highest ET of 59.97 mm in March and 76.19 mm in April. A high ET indicates a growing season, during which plants require more water. Efficient agricultural water management requires an understanding of ET. Irrigation practices need to be modified to sustain soil moisture throughout March and April, when ET increases. A higher SURFQ was observed during the wet season (June–August), peaking in July. The

LATQ peaked in August which could play a vital role in sustaining groundwater levels during the dry season, conserving stream base flow. WYLD increased in summer, July (128.61 mm), and August (150.16 mm). SYLD peaked in August (21.43 T/HA), correlating with the highest precipitation a critical for effective water management.

SSP245_S1 (Fig. 5a) indicated moderate precipitation, peaking in July and August, reaches approximately 289.49 mm and 299.32 mm, respectively. ET showed moderately high in July and August around 15.44 mm and 17.26 mm, respectively. The PET in July and August was lower than that in the later periods, suggesting that the temperature will rise in the future. Precipitation in SSP245_S2 (Fig. 5a), July (281.75 mm), and August (295.24 mm) remained high but was slightly lower than that in SSP245_S1 (Fig. 5a), reflecting rainy season variability. SURFQ and LATQ remained nearly as those in SSP245_S1 (Fig. 5a), with minor variations. SSP245_S2 showed a seasonal pattern of SSP245_S1 but with slight shifts. Precipitation increased in January, February, and throughout the rainy season (311.07 mm) in August. The SURFQ and LATQ show an increase compared to the former periods, mainly during the rainy season, when the runoff reaches August. In July and August, WYLD was slightly lower than that of SSP245_S1 (Fig. 5a). PET exceeded that of SSP245_S1 (Fig. 5a), demonstrating an increased temperature pattern which might create potential drought risk. PET (Fig. 5a) exceeded SSP245_S1 (Fig. 5a) and SSP245_S2 (Fig. 5a). SSP245_S3 (Fig. 5a) showed greater alterations, with WYLD reaching a maximum of 232.12 mm.

Considerable annual precipitation occurred throughout SSP370_S1 (Fig. 5b). In July (294.02 mm) and August (302.77 mm), maximum precipitation causes high SURFQ (98.96 mm and 101.80 mm) and LATQ (18.18 mm and 20.69 mm). August had the highest WYLD (230.38 mm due to maximum precipitation. The PET was lower, indicating a moderate but increasing trend. In SSP370_S2 (Fig. 5b), July (285.20 mm) and August (298.78 mm) experienced the highest rainfall. The precipitation, SURFQ, and LATQ decreased slightly in SSP370_S1 (Fig. 5b). In August, ET rose to 17.30 mm, whereas WYLD rose to 225.91 mm may be because plants and the atmosphere require more moisture with rising temperatures. The SSP370_S2, PET (Fig. 5b) exceeded SSP370_S1 (Fig. 5b) by 19.37 mm in February, suggesting rising temperatures increase atmospheric water loss, and SSP370_S3 (Fig. 5b) represents far-future scenario.

SURFQ and LATQ follow a similar pattern, with August showing highest values (105.39 mm and 20.56 mm). SYLD peaked in August (32.31 T/HA), indicating increased erosion risks. In August, the WYLD (229.53 mm) maintained water flow. However, ET remained high (17.21 mm in August), indicating an increase in atmospheric moisture.

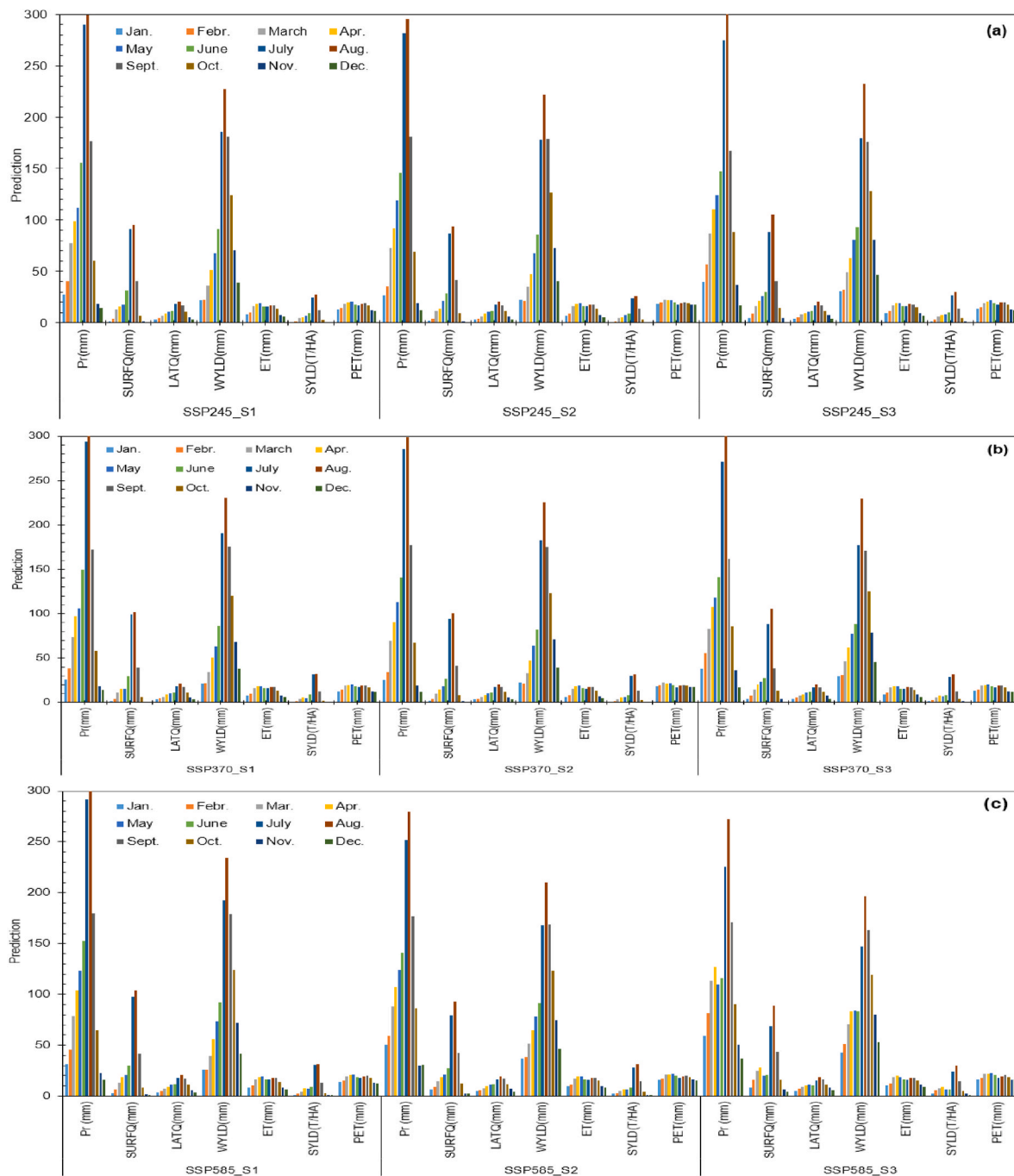


Fig. 5. Hydrological balances under: (a). SSP245, (b). SSP370, (c). SSP585.

PET peaked at 19.64 mm in April, revealing that while precipitation is rising, a warmer climate increases ET and PET, reducing the ecological and agricultural water supply. In August, SYLD had a high erosion risk due to rainfall and runoff. The SSP370 scenario demonstrated trends showing rising rainfall intensity, particularly in July and August.

SSP585_S1 (Fig. 5c) showed maximum precipitation from June to August. In July, precipitation of 292.14 mm resulted in a significant SURFQ of 98.05 mm and considerable WYLD of 192.76 mm. The SYLD showed a considerable increase, indicating a high risk of soil erosion. In SSP585_S2 (Fig. 5c), wet seasons augmented precipitation and SURFQ. WYLD was still high in SSP585_S3 (Fig. 5c), and ET rose by 16.07 mm in July owing to hot temperatures and increasing atmospheric moisture. This shows that under SSP585, an increase in precipitation and SURFQ during the wet season led to a high WYLD. Rising SYLD indicates land

degradation, impacting water quality and agriculture. ET and PET show increased evaporative demand, worsening water scarcity and requiring adaptive management during dry periods, challenging for ecological management. Analyzing these trends improves water resource management and enhances decision-making for irrigation, flood control, and ecosystem preservation.

3.6. Evolving water balances and management

Comparing projected changes in water resources to historical patterns reveals major changes in water balances. These shifts have significance for water management because they have crucial repercussions. Under Historical trends, July and August had the highest rainfall, while March and April experienced the highest of both PET and

ET, highlighting key periods for agricultural water requirements. The peak of SURFQ and LATQ occurred during the wet season, which contributed to the replenishment of groundwater and the protection of streamflow. Moreover, both WYLD and SYLD were at their maximum during the summer, coincided with the patterns of rainfall, and highlighted the significance of effectively managing soil erosion and water availability.

Future estimates under different scenarios indicate a rise in rainfall during the rainy season. WYLD exhibits some fluctuation, with certain scenarios predicting minor declines despite increasing rainfall, contrasting with SURFQ and LATQ, which continue to be remarkably high during the wet season. To effectively manage water resources, adaptive measures will be necessary. These strategies will include modifying irrigation operations to handle increased ET and PET, particularly during March and April. It is important to strengthen flood control systems to effectively manage high SURFQ and WYLD during severe rainfall periods. It is essential to make efforts to conserve soil to reduce SYLD and protect water quality. To reduce the likelihood of drought and soil erosion, future initiatives are desirable to consider shifting patterns of rainfall, rising temperatures, and an increasing need to set adaptive measures to assure long-term water security.

3.7. Watershed hydrologic vulnerability

The Hydrological Vulnerability Index (HVI) is a framework designed to assess water resource resilience by integrating numerous hydrological variables (Fig. 6). HVI provides a quantitative measure of watershed vulnerability to hydrological stress and water resource challenges. The calculated HVI ranged from low to extreme, with lower values indicating resilience to hydrological stress and higher vulnerability, as indicated in (Table 1) and (Fig. 7). In the baseline scenario (Fig. 7a), the northeastern region was the least vulnerable covering approximately 54.03 % of the low HVI (Table 1). Moderate HVI was largely concentrated in the southeastern regions, accounting for approximately 17.75 %, whereas high HVI was observed in the central regions, accounting for 11.58 %. The extreme HVI (16.64 %), covering the plains extending from west to east, highlights the relative hydrological balance in the northeastern region. Conversely, the high and extreme HVI of the central plain suggests vulnerability to water stress.

Table 1

HVI under baseline and future scenarios with areal coverage.

HVI	Areal Coverage (%)			
	Baseline	SSP245	SSP370	SSP585
Low	54.03	26.48	49.21	49.51
Moderate	17.75	2.86	32.65	18.46
High	11.58	27.21	7.16	14.01
Extreme	16.64	43.45	10.98	18.02

Under SSP245 (Table 1), the low HVI decreased to 26.48 %, more prominent in the western and eastern regions (Fig. 7b). Moderate HVI (2.86 %) primarily affected the central area (Fig. 7b). High HVI surges to 27.2 %, concentrated in central parts (Fig. 7b), whereas extreme HVI covers 43.45 %, predominantly affecting northeastern region (Fig. 7b). This change indicates a substantial increase in hydrological stress under moderate climate scenario, possibly driven by moderate warming increasing evapotranspiration rates. The SSP370 scenario (Table 1) indicates a more balanced HVI than SSP245 scenario. Low HVI (49.21 %) was concentrated in eastern and southern regions (Fig. 7c). The moderate HVI increased to 32.65 %, covering areas from north to west, reflecting balance of water stresses. The high HVI decreased to 7.16 %, confined to the eastern region, whereas extreme HVI decreased significantly to 10.98 %, affecting northeastern region. Hence, the intermediate emission pathway could mitigate extreme HVI under SSP245.

In the most severe SSP585 (Table 1), the low HVI increased slightly to 49.51 % in the eastern and western regions (Fig. 7d). Moderate HVI, distributed mainly in the northeastern and southern parts, affected 18.46 % of the regions. The high HVI increased to 14.01 %, concentrated in the northern regions, whereas the extreme HVI increased to 18.02 %, spanning the west to east plains. This suggests that although certain areas benefit from low HVI due to likely favorable hydrologic changes, regions with extreme HVI remain significant under SSP585. The high and extreme HVI regions in SSP245 highlight the need for watershed conservation and sustainable agriculture. The variation in vulnerability in SSP370 and SSP585 required flexible scenario-based water management. A moderate HVI implies intermediate vulnerability to water stress. Extreme HVI is prone to water scarcity, catastrophic drought, and flooding, which harm ecosystems and humans. Hence, this new HVI framework provides a valuable tool for integrating climate

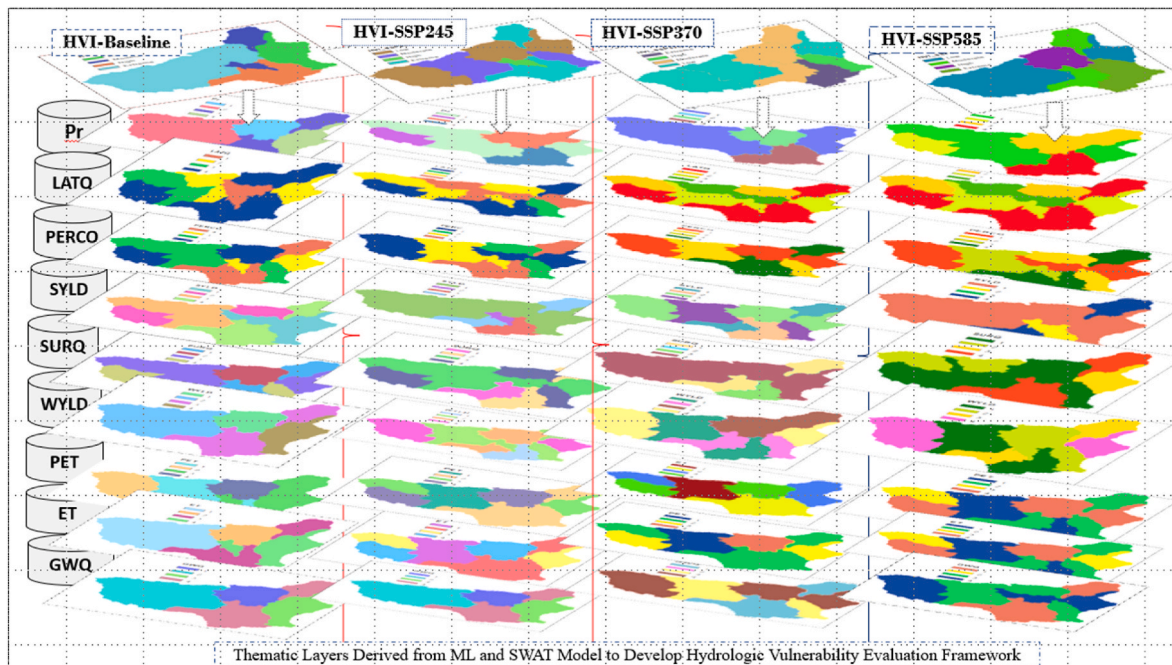


Fig. 6. Thematic layers derived used for HVI development frameworks.

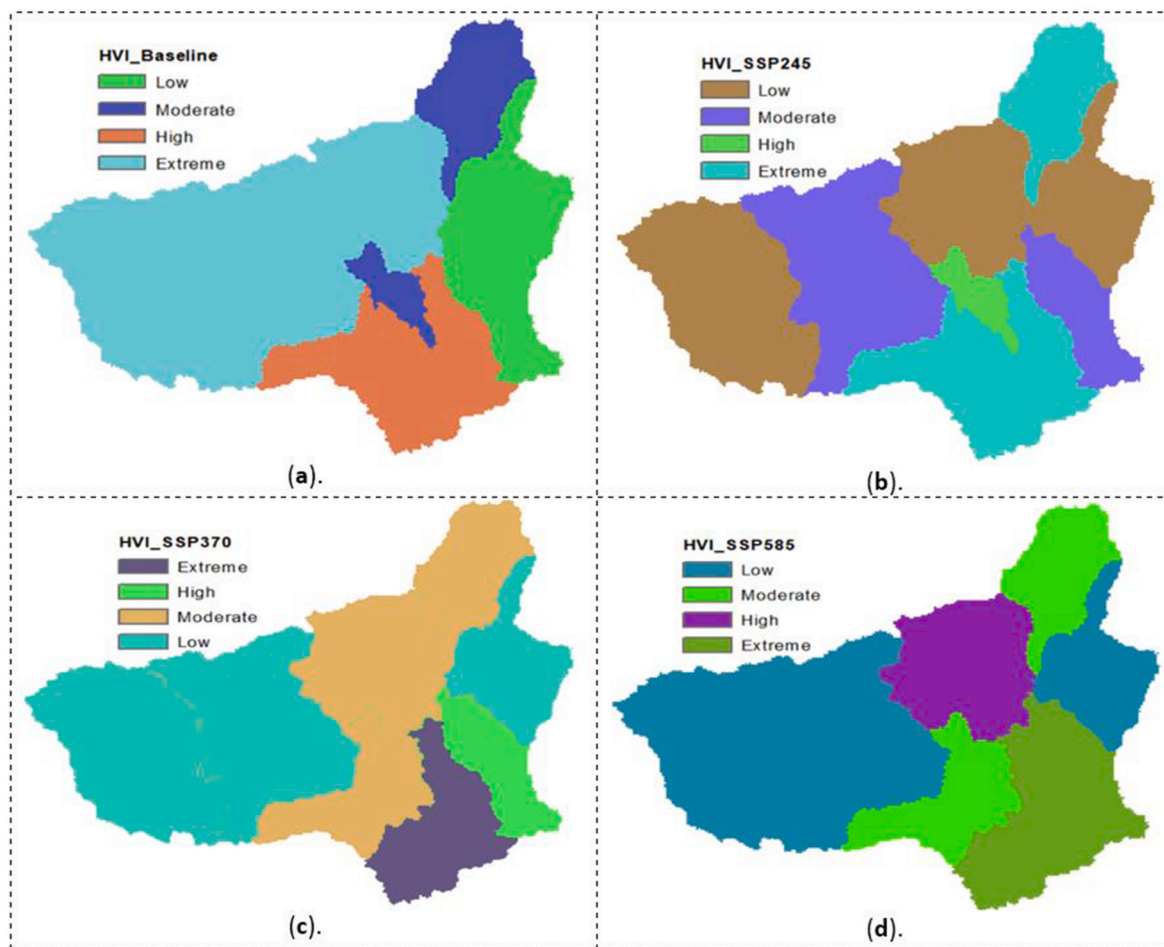


Fig. 7. Watershed (HVI) under: a) baseline, b). SSP245, c). SSP370, d). SSP585 scenarios.

responsiveness with sustainable water and environmental management.

The SSP370 scenario differs from the SSP245 scenario in terms of HVI distribution, particularly in the coverage of vulnerability levels. Under SSP370, the low vulnerability decreases from 49.21 % in SSP245 to 26.48 %, while moderate vulnerability increases meaningfully from 2.86 % to 32.65 %. Conversely, the high vulnerability decreases from 27.21 % in SSP370 to 7.16 % in SSP245, and extreme vulnerability shows a slight increase, from 10.98 % to 43.45 % in SSP370. This alteration proposes that, under SSP370, moderate and extreme HVI are more widespread compared to SSP245. While low vulnerability drops, additional areas experience higher levels of vulnerability, indicating increased water stress. This indicates that under SSP370, challenges related to water availability are likely to increase, with a larger portion of the area moving into higher vulnerability. In contrast, the SSP245 scenario, which assumes lower emissions, shows less severe changes toward higher vulnerability. The shift from SSP245 to SSP370 suggests a larger risk of water scarcity and stress in the later scenario.

The SSP585 scenario shows distinct differences in HVI distribution compared to SSP245. Under SSP585, the "Low" HVI decreases slightly compared to SSP245, from 49.21 % to 49.51 %, but the "Extreme" HVI increases substantially, from 43.45 % to 18.02 %. This suggests that under higher emissions scenarios, more regions experience extreme levels of hydrological vulnerability, leading to keen water stress. In contrast, the SSP245 scenario shows a more balanced distribution of HVI across categories, with less extreme water stress and more areas in "Low" and "Moderate" classes. This indicates that under moderate emissions, water stress is less severe, and with lower hydrological vulnerability remain more widespread. Overall, the comparison suggests that higher

emissions scenarios like SSP585 are likely to result in more regions facing extreme water stress, while SSP245 could help maintain a broader range of with lower vulnerability. Therefore, reducing emissions and transitioning toward more moderate pathways reduce pressures on water resources and reduce the extent of extreme water stress.

3.8. Implications for climate responsive water management

The moderate SSP245 emission scenario is expected to peak at approximately 2040 and then decline, owing to a slight increase in precipitation during the wet season. The increased precipitation intensity in the late SSP585 scenario indicates a larger uncertainty in the hydrological cycle. PERCO and LATQ reflect groundwater recharge and subsurface flow, respectively, where a higher reduces vulnerability and supports water resilience (Scanlon et al., 2023; Taylor et al., 2013). High SURQ aggravates erosion, flood risk, and water resource stress. SSP585 had highest SYLD in the late future SSP585, indicates a high risk of soil erosion and land degradation. A larger recharge is indicated by a higher PERCO, resulting in pollutant leaching in the urban and agricultural regions. Although lower values may suggest that less water is available downstream, a higher LATQ implies higher SURQ. SYLD and SURQ indicate erosion and flooding risks, respectively, with higher values exacerbating stress associated with ecosystem health and water quality (Pimentel et al., 1995). This approach highlights spatial variations and provides actionable insights into targeted interventions in various scenarios (IPCC, 2021). High sediment loads diminish reservoir capacity and lower water quality, thereby increasing the cost of water treatment and ecosystem restoration. This divergence indicates that SSP585 is

associated with extreme rainfall events that affect the water availability.

To mitigate the rising risk of soil erosion under various climate scenarios, a combination of sustainable land management, vegetation conservation, and hydrologic interventions is crucial. Conservation practices such as no-till farming, contour plowing, and agroforestry reduce soil loss and regulate surface runoff, preventing excessive sediment transport. Maintaining ground cover through afforestation, cover cropping, and riparian buffers enhances infiltration, stabilizes soil, and minimizes erosion caused by wind and water. Structural measures like terracing, check dams, and sediment retention basins further control sediment flow and reduce downstream impacts. Additionally, strengthening climate-adaptive policies, improving soil health with organic matter, and promoting responsible land-use planning enhance watershed resilience. These integrated approaches collectively safeguard soil stability, preserve water quality, and mitigate long-term hydrological vulnerabilities associated with climate-induced erosion.

SSP370 had a slightly higher PET than SSP245, reflecting a more cautious change in climate estimates that affects land and water management. In runoff-prone agricultural and urban areas, the SSP585 requires soil erosion management. SSP585 and SSP370 had higher PET values, indicating a need for better water conservation and irrigation. The demand for water rises with ET, raising the possibility of droughts, decreasing agricultural production, and putting stress on ecology. A higher PET indicates greater water stress, vulnerability to drought, reduced agricultural output, and ecosystem decline. ET and PET represent water loss and atmospheric demand, respectively, with high values indicating increased water stress, and low values indicating sufficient water availability.

Groundwater replenishment could be improved by managing WYLD of SSP370 and SSP585. A higher WYLD can lead to management issues, even if it indicates a supplementary water supply. While a lower GWQ indicates vulnerability to water scarcity, a higher GWQ indicates better resilience, thus, supporting water needs. WYLD and GWQ represent the water availability. Low values indicate scarcity and high vulnerability, whereas high values indicate better resilience (Arnold et al., 1998). Precipitation is the primary source of water input, with low levels increasing vulnerability, whereas extreme rainfall can increase the flood risk (IPCC, 2021). Under SSP245, an extreme HVI requires adaptive infrastructure, water storage capacity, and policies to mitigate hydrological regime changes. Under SSP585, rising low HVI and reducing extreme HVIs require advanced action. SSP245 suggests a more manageable future with modest water stress, whereas SSP370 and SSP585 show more serious consequences for higher emission scenarios.

Hence, to mitigate flood risks from increased summer precipitation while ensuring water storage for dry seasons, a range of integrated water management strategies could be employed. Flood retention basins or detention ponds can be constructed to capture excess water during heavy rainfall, releasing it gradually to prevent downstream flooding while providing storage for dry periods. Rainwater harvesting and infiltration techniques, such as permeable pavements, and green infrastructure, can help manage runoff and retain water. Furthermore, land management planning and floodplain mapping can limit development in flood-prone areas, reducing vulnerability. Managed aquifer recharge systems, can store water underground for later use. Smart reservoir management strategies that balance flood control with water storage for droughts can optimize water availability ensuring sustainable water management while reducing flood risks.

4. Conclusions

This study used machine learning and the SWAT model to examine water availability under bias-corrected CMIP6 GCMs for the SSPs. ERT and CatBoost were found to be the most effective machine-learning algorithms. The ensemble SWAT model calibration demonstrated strong agreement between simulated and observed values, supported by statistical indicators that confirmed reliability and reduced input

uncertainties using bias-corrected datasets. This improves the accuracy of hydrological vulnerability estimates. Moreover, we developed a novel HVI to quantify spatiotemporal hydrological dynamics to watershed vulnerability. By linking vulnerability to climatic forcing, the HVI could provide water resilience assessment framework. Water resource management should address excess water during floods and scarcity during droughts as hydrological cycles intensify. This study emphasises the importance of climate projections in water management strategies, with the aim of establishing a framework for adaptive management that benefits ecosystems.

The findings highlight substantial spatial variability in hydrological stress, emphasizing the need of scenario-specific water management strategies. Moderate HVI reflects intermediate vulnerability, while extreme HVI denotes sensitive risks of water scarcity, drought, and flooding, with severe implications for ecosystems and communities. Extreme rainfall events under SSP585 pose additional challenges, such as soil erosion, land degradation, and increased water treatment costs. Effective water conservation measures and adaptive infrastructure are essential to mitigate these risks. Furthermore, increased atmospheric water demand under SSP370 and SSP585 raises the potential for drought, threatening agricultural productivity and ecological health. Precipitation patterns under SSP245 suggest manageable water stress, while SSP370 and SSP585 reveal greater challenges from higher emissions, including extreme rainfall and associated flood risks.

The current study has limitations regarding the explicit quantification of uncertainty in model predictions. Therefore, developing and integrated probabilistic frameworks into machine learning would enhance the reliability of climate change projections and support more robust decision-making for policy, water management, and risk assessments. Moreover, advanced ensemble machine learning can be coupled with aquifer systems to synergistically improve water management and provide insights.

CRedit authorship contribution statement

Tarekegn Dejen Mengistu: Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Visualization, Software, Validation, Data curation, Investigation, Formal analysis. **Sun Woo Chang:** Writing – review & editing, Validation, Resources, Visualization. **Il-Moon Chung:** Writing – review & editing, Resources, Project administration, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2025.126025>.

Data availability

Data will be made available on request.

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